CASE STUDY / TURNKEY INFRASTRUCTURE

Plant connectivity. Integrated visibility.

DCM Shriram Limited
Shriram Alkali & Chemicals | Jhagadia

Cisco networking, fibre infrastructure, surveillance licensing and storage, industrial connectivity and a Dell-specified server platform.

CONSOLIDATED PO VALUE | APPROX. USD 26,533

13 supplied POs | Includes GST | Indicative conversion INR 95.36 per USD; reported close, 13 October 2026.



Executive brief | Ordered scope translated into one design

The 13 supplied orders place D-TECH at the intersection of plant networking, CCTV expansion, industrial communication and server supply. This document translates those materials and the coal-yard installation order into an understandable integration design. The POs establish ordered scope; as-built drawings and completion records were not supplied.

18
CISCO SWITCHES

65 + 65
VSS / PSTOR CHANNEL LICENCES

130 TB
RAW SURVEILLANCE DISKS

13
UNIQUE PURCHASE ORDERS

INTEGRATION OBJECTIVE

Link distributed access racks through tested fibre, carry camera traffic to approved recording services, protect permitted connections and document the server environment.

DESIGN BOUNDARY

Existing cameras, recorder / VMS servers, aggregation nodes, backup targets and endpoint allocation must be surveyed. Material counts are not proof of installed capacity.

Pages 03-04 PO analysis | 05 Material map | 06 Network drawing | 07 Security / industrial paths | 08 CCTV / storage | 09 Server | 10 Delivery | 11 Acceptance | 12 Handover

PO analysis | Execution, switching and passive materials

PO	Order date	Basic INR	Gross INR	Ordered material / job contribution
4250021274	14.08.2025	40,325.00	47,583.50	Coal-yard execution: 600 m fibre laying; 915 m STP laying; 5 LIUs; 84 fibre splices and 84 OTDR tests; 4 L2 switch configurations; 13 camera setups; 12 brackets; 1 DVR/NVR setup; 3 rack fixing/dressing; RJ45, tags and terminations.
4250021428	29.08.2025	45,500.00	53,690.00	FortiGate 40F line with FortiCare Elite support SKU FC-10-0040F-284-02-36; quantity 1. Hardware versus support entitlement must be reconciled from the approved offer.
4210029249	02.09.2025	201,443.00	237,702.74	1,400 m 12-core SM fibre; 610 m outdoor Cat6; 1 x 24F + 3 x 12F LIUs; 7 LC-LC cords; 6 patch panels; 86 IOs; 63 x 1 m + 28 x 2 m cords; 100 RJ45s; 4 managers; casing, boxes and faceplates.
4250021459	02.09.2025	359,340.00	424,021.20	4 Cisco C1300-8P-2G + 4 Cisco 24-port PoE switches; 5 x 10 TB surveillance HDDs. Confirm exact 24-port SKU / uplink specification.
4210029301	05.09.2025	10,654.00	12,571.72	1 x 305 m Commscope Cat6 drum; 20 x 1 m patch cords. Separate PO, not a duplicate of the earlier cable package.
4250021518	09.09.2025	640,615.00	755,925.70	6 C1300-48P-4G; 4 C1300-8P-2G; 8 MGBLH1 SFPs; 15 VSS + 15 pStor licences; 150 m fibre; 1,220 m Cat6; LIUs / cords / IOs; outdoor rack; 15 camera stands; boxes, glands, SS fixings; 8 x 10 TB HDDs.
4250021602	17.09.2025	142,244.00	167,847.92	1,500 m 12-core SM fibre; 20 IOs; 20 x 1 m patch cords; 4 x 24F LIUs; 8 LC-LC duplex cords.

Gross includes CGST + SGST. Dates use the first-page Order Date; some repeated page headers show later dates. Each PO is counted once.

PO analysis | Licensing, industrial links and compute

PO	Order date	Basic INR	Gross INR	Ordered material / job contribution
4250021646	22.09.2025	140,250.00	165,495.00	50 HikCentral VSS channel licences + 50 pStor channel licences, including licence integration in HikCentral.
4210029639	24.09.2025	78,200.00	92,276.00	2 Moxa EDS-208A-S-SC unmanaged industrial switches: 7 copper 10/100 ports + 1 single-mode SC 100BaseFX port each; DIN rail; dual DC inputs.
4250021668	24.09.2025	10,550.00	12,449.00	2 Netgear AGM732F single-mode 1310 nm LC/LX optics. Validate approved host switches, line rate and peer compatibility.
4250021693	27.09.2025	27,000.00	31,860.00	1,000 m 6-core armoured optical fibre. Endpoints and termination allocation to be confirmed.
4250021834	16.10.2025	49,875.00	58,852.50	1 enclosure for an existing 55-inch display; GI / aluminium composite construction, toughened glass, vents and fan; existing swivel mount; one-time installation included.
4250022503	27.12.2025	398,245.00	469,929.10	1 Dell-specified R2 server machine: Xeon E-2468, 16 GB, 2 x 960 GB SATA SSD, RAID 1, single 600 W PSU, Windows Server 2022 Standard or latest; 3-year OEM warranty. Generic ML30 label conflicts with Dell wording.

Reconciled totals: basic INR 2,144,241.00 + GST INR 385,963.38 = gross INR 2,530,204.38

Cover conversion: gross INR total ÷ 95.36, rounded to whole dollars. PTI / ThePrint reports the 13 October 2026 close; this is indicative, not historical invoice FX.

Only the 13 attached POs are included. Referenced earlier orders are not added; PO values are not certified billing, delivery or completed-work values.

Material map | What the ordered components enable

Material group	Consolidated ordered baseline	Integration purpose
FIBRE BACKBONE	3,050 m 12-core SM armoured fibre + 1,000 m 6-core armoured fibre; 7 x 24F LIUs + 3 x 12F LIUs; 25 LC-LC duplex cords.	Rack-to-rack backbone, terminated core map, optical test records and labelled patching.
COPPER / PASSIVES	1,830 m outdoor Cat6 + 1 x 305 m normal drum; 8 x 24-port unloaded panels; 122 IOs; 119 x 1 m + 28 x 2 m cords; 200 RJ45s.	Local endpoint / rack connectivity; quantities need outlet and port allocation.
RACK / CONTAINMENT	1 outdoor 9U rack; 12 cable managers; 50 m casing; 25 m flexible conduit; 43 back boxes; 25 single + 33 dual faceplates.	Cable protection, patch organisation and maintainable termination. Service order separately covers 3 rack installations.
CAMERA MECHANICALS	15 fabricated stands; 9 junction boxes; 9 x 25 mm + 6 x 19 mm glands; 60 SS fixings.	Field mounting / cable entry; confirm weather, corrosion and hazardous-area requirements for each location.
ACTIVE NETWORK	18 Cisco switches: 8 x 8-port, 4 x 24-port, 6 x 48-port; 8 Cisco + 2 Netgear optics; 2 Moxa industrial switches.	Distributed PoE access and fibre links; peer optics and speeds must be compatible.
VIDEO / COMPUTE	65 VSS + 65 pStor channel licences; 13 x 10 TB HDDs; 1 server; 1 firewall/support line; 1 existing-display enclosure.	Licensed surveillance expansion, recording media and compute; role / disk allocation require an approved design.

Service quantities: 13 camera setups, 4 switch configurations and 1 DVR/NVR setup belong to the coal-yard job, not the full physical inventory.

Network architecture | Fibre backbone, distributed access

APPROVED EXISTING LAN

Customer aggregation / routing point; topology, VLANs and uplink capacity to be verified.

FIBRE PATCH / AGGREGATION

SM backbone → LIU → LC patch → approved SFP peer. Match speed, wavelength, loss and host support.

OPERATIONS SERVICES

HikCentral / recording and client access on permitted segments; existing hosts to be identified.

48-PORT ACCESS | 6 UNITS

C1300-48P-4G; local PoE endpoints; four 1G SFP uplinks per ordered specification.

24-PORT ACCESS | 4 UNITS

Cisco managed PoE+; exact SKU and uplink mapping to be confirmed.

8-PORT ACCESS | 8 UNITS

C1300-8P-2G; distributed field endpoints; select approved combo-uplink mode.

Switch copper-port nameplate total: 448. This is not available camera capacity: uplinks, PoE budget, segmentation and other endpoints reduce usable allocation.

Proposed star / hierarchical concept. Physical routes, rack positions, redundancy and bandwidth must be validated on site; no 10G backbone is ordered here.

Security and industrial links | Two distinct service paths

PERMITTED EXTERNAL PATH

Existing WAN / upstream service. Customer defines whether this connection serves IT or an approved remote service.

FORTIGATE 40F LINE

40F + FortiCare Elite wording; validate hardware delivery and support scope. Security subscriptions are not assumed.

APPROVED INTERNAL ZONE

Allow only required flows; no default unrestricted path into plant control or CCTV administration.

INDUSTRIAL COPPER

Existing field devices; 7 x 10/100BaseT(X) ports per Moxa EDS-208A-S-SC.

MOXA EDGE | 2 UNITS

Unmanaged, DIN rail, dual DC input. 100BaseFX single-mode SC optical port; upstream VLAN control is separate.

MATCHED 100BASEFX PEER

Validate optical peer, connector / patch path and power supply. Do not connect by assuming 1G SFP equivalence.

The 100 Mbps Moxa optical interface is distinct from the ordered 1G Cisco / Netgear optics. No managed ring, automatic redundancy or P2P radio is inferred.

Surveillance architecture | Live view, recording and evidence

FIELD CAMERAS

Existing / customer-approved camera inventory. Coal-yard service order covers 13 installation / setup tasks.

CISCO ACCESS → VMS

Approved camera VLAN / routing path to HikCentral; 15 + 50 VSS channel licences ordered.

USER-WISE ACCESS

Proposed least-privilege live view, playback and export by approved role; final users / policies to be confirmed.

MEDIA BASELINE

13 x 10 TB = 130 TB raw aggregate. Ordered across 2 POs; not one confirmed array or usable volume.

RECORDING / PSTOR

15 + 50 recording channel licences. Confirm recorder architecture, host sizing, disk bays and supported storage design.

PLAYBACK / EXPORT

Test recorded footage, time alignment and an authorised export; record results with the security owner.

Illustrative sizing only: 65 channels × 4 Mbps × 86,400 s ÷ 8 = 2.808 TB/day; 30 days = 84.24 TB before overhead / RAID. These settings are not confirmed.

Raw 130 TB does not guarantee 30-day retention. Validate bitrate, recording duty cycle, usable RAID capacity, licences, throughput and application host limits.

Server architecture | Mirrored boot / data platform

ORDERED SERVER | 1 UNIT

WINDOWS SERVER
2022 Standard or latest; validate edition, licensing and workload support.

COMPUTE RESOURCES
Xeon E-2468 8C/16T; 16 GB UDIMM; actual service workload sizing required.

STORAGE / POWER
2 x 960 GB SATA SSD; RAID 1; single 600 W PSU; 3-year OEM warranty.

RAID 1 | TWO-DISK MIRROR

SSD A 960 GB ↔ SSD B 960 GB
Approx. 960 GB decimal before formatting / OS. Disk mirroring protects against a drive failure; it is not backup.

INDEPENDENT RECOVERY PATH

Backup repository and backup software are not established by these POs. Define protected data, schedule, restore tests, ownership and RPO / RTO.

Naming conflict: PO says ML30 Gen10 Plus but also “Make: Dell only” / R2 server and Dell SSDs. Obtain the approved final OEM model and configuration.

Do not assume this server is the HikCentral/pStor host. One 16 GB server with a single PSU does not establish an HA, VM cluster or disaster-recovery solution.

Implementation | Material to commissioned service

01 | SURVEY & FREEZE

Map racks, routes, existing cameras, licences, server role and display. Resolve model / optical conflicts.

02 | RECEIVE & VERIFY

Reconcile each PO; check part / serial / licence identity, quantities, condition and approved documents.

03 | INSTALL PASSIVES

Route / label cable; mount approved racks and stands; terminate, splice and test fibre / copper.

04 | CONFIGURE ACTIVE

Approve VLAN / IP plan; configure switches / firewall, peer optics, permitted flows and secured management.

05 | INTEGRATE SERVICES

Set up approved cameras / recorder; integrate VSS / pStor licences; configure server RAID / OS and enclosure.

06 | TEST & HAND OVER

Validate network / PoE, recording, storage, access and recovery; submit as-built records and sign-off.

Documented execution: 600 m fibre, 915 m STP, 84 splices / OTDR tests, 13 camera setups, 4 switch configurations and 1 recorder setup. Other tasks require scope agreement.

Engineering review | Acceptance gates and design exceptions

Gate / issue	Required engineering action	Acceptance evidence
OPTICAL COMPATIBILITY	Separate 100BaseFX SC Moxa links from 1G LC/LH/LX links; approve host/SFP support, wavelength, connector and measured optical loss.	Matched peer list, core / port map, test results and error-free link observation.
POE / PORT BUDGET	Use nameplate port totals only as a starting point. Calculate worst-case endpoint power per switch and uplink bandwidth.	PoE allocation sheet; representative load / packet-drop test.
LICENCES / DEVICE COUNTS	Reconcile 65 VSS + 65 pStor licences against inventory; distinguish 13 setup tasks and 15 stands from total cameras.	Licence entitlement / activation records, channel mapping and live-view tests.
RECORDING CAPACITY	Map 13 disks to actual recorder bays, supported disks and usable RAID layout. Validate throughput and retention at configured bitrates.	Disk health, usable volume, earliest playback and authorised export results.
SERVER / FIREWALL SCOPE	Resolve Dell/ML30 naming; confirm final host role and OS rights. Clarify 40F hardware versus support SKU; approve firewall policy.	Approved BOM, serials, licence / support records and permitted-flow tests.
PHYSICAL / SAFETY	Confirm cable routes and lengths; installation rack count differs from supplied rack count; enclosure is for an existing 55-inch screen.	As-built routing, OTDR / copper tests, rack / enclosure inspection and site acceptance.

No published uptime, response-time, cybersecurity certification, retention guarantee or job-completion outcome is inferred from purchase orders.

D-TECH delivery value | One accountable integration record

DESIGN & TRACEABILITY

PO-to-material map; approved topology, rack / port allocation, camera register and resolved engineering exceptions.

TESTED IMPLEMENTATION

Fibre / copper records, active configuration baseline, licence integration checks and service validation evidence.

SUPPORT READINESS

Asset / serial register, OEM entitlement, backup ownership, operator handover and clear issue escalation.



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D-TECH | CUSTOMER CARE

Open support portal

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Basis: 13 supplied POs, August-December 2025, and customer visuals. This is ordered-scope analysis and a reference design, not a completion certificate.

References: Cisco Catalyst 1300 data sheet | Moxa EDS-208A-S-SC | PTI / ThePrint FX close, 13 Oct 2026